American River College – Math 372
Spring 2020 – Class 24/32

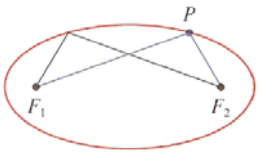
Instructor:	John Burke
E-mail:	burkej@arc.losrios.edu
Telephone:	916-484-8601
Class:	A & S 102, TTh 8:15 – 10:20
Student hours:	Howard Hall 148, MW 1:00 – 2:30, TTh 1:00 – 2:00

Today:	§§ 11.2; Homework #5 Due
Assignment:	Online Homework #11
Next:	§§ 11.3/11.4

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11.2 Ellipses

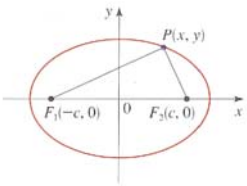
Definition: An **ellipse** is the set of all points in the plane the sum of whose distances from two fixed points $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$ (the foci) is a constant.



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11.2 Ellipses

To obtain the simplest equation for an ellipse, we place the foci on the x-axis at $\boldsymbol{F}_1(-c, 0)$ and $\boldsymbol{F}_2(c, 0)$, so the origin is halfway between them. Let the sum of the distances from a point on the ellipse to the foci be $2a$. If $\boldsymbol{P}(x, y)$ is any point on the ellipse, we have $d(\boldsymbol{P}, \boldsymbol{F}_1) + d(\boldsymbol{P}, \boldsymbol{F}_2) = 2a$.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad \text{with } a > b$$

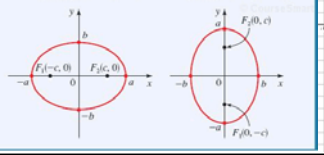
$$c^2 = a^2 - b^2$$

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11.2 Ellipses

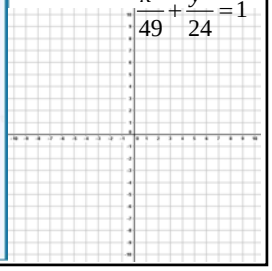
ELLIPSE WITH CENTER AT THE ORIGIN

The graph of each of the following equations is an ellipse with center at the origin and having the given properties.

EQUATION	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$	$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$
	$a>b>0$	$a>b>0$
VERTICES	$(\pm a, 0)$	$(0, \pm a)$
MAJOR AXIS	Horizontal, length $2a$	Vertical, length $2a$
MINOR AXIS	Vertical, length $2b$	Horizontal, length $2b$
FOCI	$(\pm c, 0),\ c^2=a^2-b^2$	$(0, \pm c),\ c^2=a^2-b^2$
GRAPH		

Find the foci, vertices, and the lengths of the major and minor axes, and sketch the graph.

$$\frac{x^2}{49}+\frac{y^2}{24}=1$$

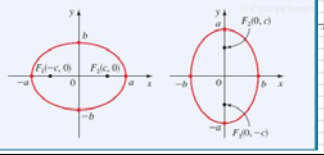


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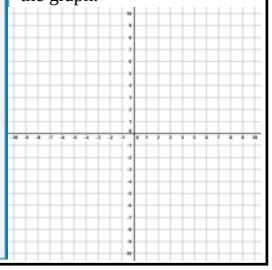
11.2 Ellipses

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GRAPH		

The vertices of an ellipse are $(\pm 4, 0)$ and the foci are $(\pm 2, 0)$. Find its equation and sketch the graph.

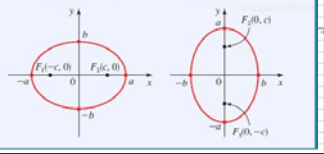


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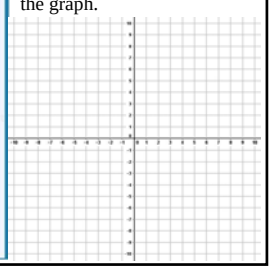
11.2 Ellipses – Your Turn

ELLIPSE WITH CENTER AT THE ORIGIN

The graph of each of the following equations is an ellipse with center at the origin and having the given properties.

EQUATION	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$	$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$
	$a>b>0$	$a>b>0$
VERTICES	$(\pm a, 0)$	$(0, \pm a)$
MAJOR AXIS	Horizontal, length $2a$	Vertical, length $2a$
MINOR AXIS	Vertical, length $2b$	Horizontal, length $2b$
FOCI	$(\pm c, 0),\ c^2=a^2-b^2$	$(0, \pm c),\ c^2=a^2-b^2$
GRAPH		

The vertices of an ellipse are $(0, \pm 5)$ and the foci are $(0, \pm 3)$. Find its equation and sketch the graph.



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11.2 Ellipses

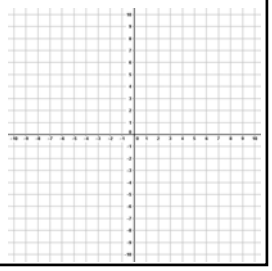
If $2a$ is only slightly greater than $2c$, the ellipse is long and thin. If $2a$ is much greater than $2c$, the ellipse is almost circular. We measure the deviation of an ellipse from circular by the ratio of a and c and call it the **eccentricity**.

DEFINITION OF ECCENTRICITY

For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ or $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ (with $a > b > 0$), the **eccentricity** e is the number

$$e = \frac{c}{a}$$

where $c = \sqrt{a^2 - b^2}$. The eccentricity of every ellipse satisfies $0 < e < 1$.



Find the equation of the ellipse with foci (±8, 0) and eccentricity e = 4/5.

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11.2 Ellipses – Your Turn

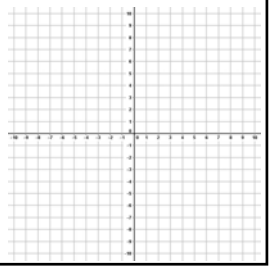
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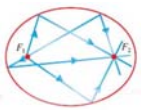


Find the equation of the ellipse with foci (±1.5, 0) and eccentricity e = .75.

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11.2 Ellipses

Ellipses, like parabolas, have an interesting *reflection property* that leads to a number of practical applications. If a light source is placed at one focus of a reflecting surface with elliptical cross sections, then all the light will be reflected off the surface to the other focus, as shown in Figure 10. This principle, which works for sound waves as well as for light, is used in *lithotripsy*, a treatment for kidney stones. The patient is placed in a tub of water with elliptical cross sections in such a way that the kidney stone is accurately located at one focus. High-intensity sound waves generated at the other focus are reflected to the stone and destroy it with minimal damage to surrounding tissue. The patient is spared the trauma of surgery and recovers within days instead of weeks.



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11.2 Ellipses

The reflection property of ellipses is also used in the construction of *whispering galleries*. Sound coming from one focus bounces off the walls and ceiling of an elliptical room and passes through the other focus. In these rooms even quiet whispers spoken at one focus can be heard clearly at the other. Famous whispering galleries include the National Statuary Hall of the U.S. Capitol in Washington, D.C. (see page 776), and the Mormon Tabernacle in Salt Lake City, Utah.



Ceiling of Statuary Hall in the U.S. Capitol (ellipse)
Architect of the Capitol

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